

Exploration of Acoustic Body Signals and Their Detectability Using 3D-Printed Ferroelectret Sensors

A Study on Signal Origins, Frequency Ranges, and Comparison with Existing Measurement Approaches

Report in the department of Electrical Engineering and Information Technology by Niels Haßler
Date of submission: April 21, 2026

1. Review: Stephan Schaumann
 2. Review: Julian Seiler
- Darmstadt



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Electrical Engineering and
Information Technology
Department

Measurement and Sensor
Technology Group

Contents

1	Introduction	3
2	Origins of Acoustic Body Signals	4
2.1	Cardiac and Vascular Sounds	4
2.2	Gastrointestinal Sounds	5
2.3	Respiratory Sounds	6
2.4	Joints and Tendons	7
2.5	Muscle Sounds	8
2.6	Overview	8
3	Sensor Types	10
3.1	Low Frequency Sensors	10
3.1.1	Seismographic Sensors	10
3.2	High Frequency Sensors	10
3.2.1	Piezoelectric Sensors	10
3.3	Ferroelectret Sensors	10
4	Discussion	11
5	Conclusion	12

1 Introduction

The human body emits acoustic signals that can provide valuable insights into a vast array of physiological and pathological processes. Ranging from the rhythmic beating of the heart to subtle sounds produced by the respiratory system and gastrointestinal tract or high frequency hits in joints, these signals offer a non-invasive window into the body's inner workings. Auscultation as one of the oldest diagnostic techniques, has been used since the days of Hippocrates (c.460 - c.370 BC) to assess the health of patients by listening to their internal sounds [1]. With the advent of modern sensor technology and advanced signal processing techniques the analysis of acoustic body signals has exceeded far beyond the capabilities of the traditional stethoscope and limited human hearing.

Differentiation of Acoustic body signals from force myography and mechanomyography This work explores the origins of various acoustic body signals and their respective spectrums. Providing excerpts of the physiological processes that generate these sounds and how they can be used for diagnostic analysis. In the second part of this work, an overview of in use and emerging sensor technologies for acoustic emissions is given, focusing on their working principles and frequency ranges. Lastly, the detectability of acoustically emitted signals using 3D-printed ferroelectret sensors is discussed and compared to existing measurement approaches.

2 Origins of Acoustic Body Signals

In this chapter the origins of acoustic body signals are explored. A focus on cardiac and vascular sounds is given first. Next gastrointestinal and respiratory sounds are discussed. Finally, acoustic emissions from joints, tendons and muscles are covered. For all signal types, the physiological processes that generate these sounds are described and characterised by their frequency content. If research exists on diagnostic capabilities or pathophysiological alterations of these signals it is covered in the respective sections.

2.1 Cardiac and Vascular Sounds

In a regular cardiac cycle the human heart emits sounds during two dominant events named the First Heart Sound (S1) and Second Heart Sound (S2) [2]. S1 is generated by the closure of the mitral and tricuspid valves at the onset of systole, while S2 stems from the halting of the aortic and pulmonary valves forcing a sudden deceleration of blood flow in the aorta and pulmonary artery, at the end of systole [2], [3]. Both actions form vibrations on the entire cardiac structure and surrounding tissues, which can then be detected as acoustic signals on the chest wall. Clinical practice has revealed a total of 5 positions on the chest wall where sounds can be best auscultated.

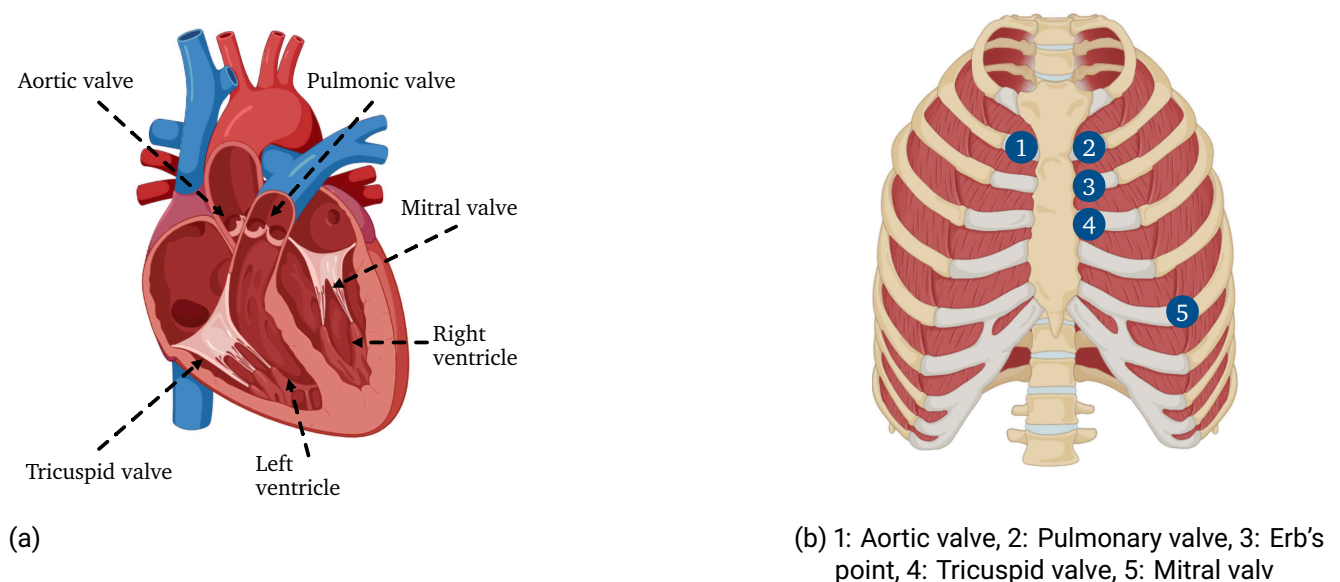
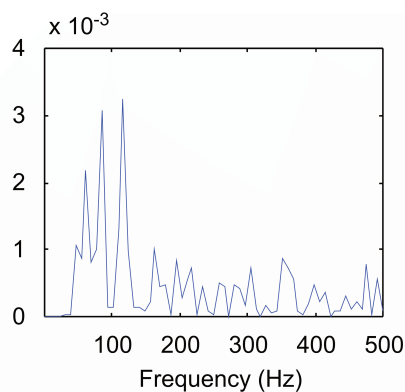
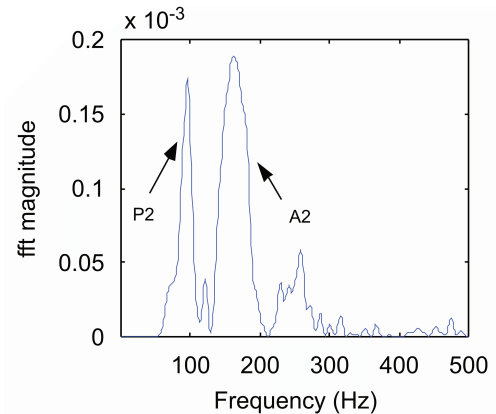


Figure 2.1: (a) Valves of the human heart and (b) Cardiac auscultation points on the chest wall.
Icons from biorender.com

Using Phonocardiography (PCG) the acoustic signals of the heart can be recorded and analyzed for diagnostic purposes. In the time-domain, S1 correspond to the QRS complex of the electrocardiogram (ECG) and S2 to the T-wave. A normal PCG signal has a frequency content in the range of 40 Hz to 200 Hz [3]. S1 typically lasts around 50 ms to 100 ms with a power spectrum in the range of 50 Hz to 150 Hz. S2 is shorter, lasting around 25 ms to 50 ms and has a higher frequency content in the range of 50 Hz to 200 Hz, which is expected as less blood is present in the cardiac chambers.



(a) Frequency spectrum for the sound S1



(b) Frequency spectrum for the sound S2

Figure 2.2: Graphs from S.M.Debbal et al. [2], (b) shows the spectral decomposition of the sound S2, where the first peak corresponds to the aortic valve and the second to the pulmonary valve.

The spectral shape of S1 and S2 are influenced by body surface area, sex and blood pressure, ventricular mass and thus may change due to disease [3], [4]. Furthermore in pathological conditions additional heart sounds may be observed, such as the Third and Fourth Heart Sound (S3 and S4) or high frequency murmurs ranging from 600 Hz to 1000 Hz [5].

Diseases in the vascular system can too produce acoustic signals. Turbulent blood flow in arteries caused by stenosis, aneurysms or arteriovenous malformations cause high pressure fluctuations in the post-stenotic region. Through the interaction of the vessel wall with the fluctuating blood flow, vibrations are generated and transmitted to the body surface [6]. The frequency content of these sounds is typically in the range of 100 Hz to 300 Hz, more severe stenosis (>70%) can produce sounds up to 1000 Hz [7].

2.2 Gastrointestinal Sounds

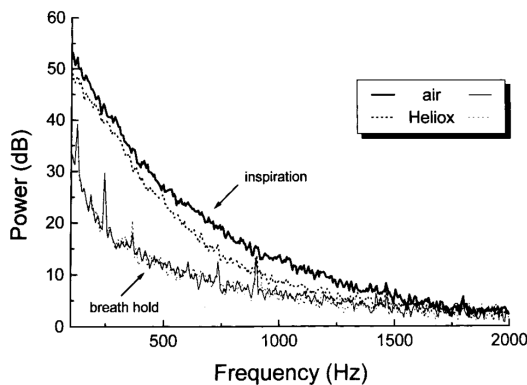
Phonoenterography (PEN) is the recording and analysis of acoustic signals produced by the gastrointestinal (GI) tract. Healthy PEN signals consist of a continuous isolated and short events in a frequency range of up to 1500 Hz with the majority of energy located between 50 Hz to 250 Hz [8].

Two approaches exist for interpreting PEN signals. The first is based on the analysis of the individual sounds and their categorisation [9]. The second approach is based on the analysis of the overall acoustic activity, like mean sound duration, silence between sounds duration, signal energy etc. The weighted combination of these features form the Acoustic Activity Index (AAI) [8]. Using the AAI, the digestive activity of the GI tract can be assessed and thus used for diagnostic purposes and long term monitoring.

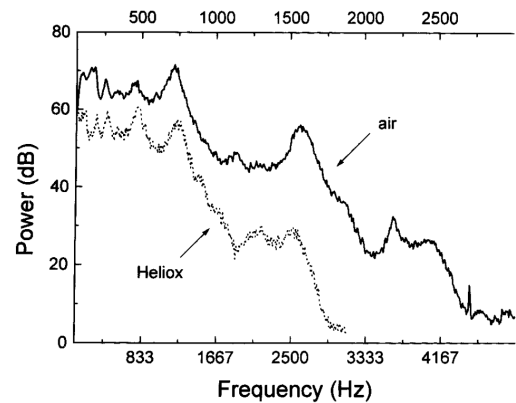
2.3 Respiratory Sounds

The respiratory cycle consist of two main phases - inspiration and expiration. Normal adults breathe 16-20 Times per minute. A healthy lung produces sounds during both phases. In inspiration sound is generated through turbulent flow in the lobar and segmental airways, in expiration sound arises from more central airways. Frequency ranges from 50 Hz to 2500 Hz [10], with the majority of energy in the range of 50 Hz to 500 Hz[11].

When deciding on sensor position for the detection of respiratory sounds, it is important to consider that the primary pathway for transmitted sound changes with frequency [11]. Lung parenchyma and tissue act as a low-pass filter and direct coupling to the chest wall is dominant. Thus transmission below 300Hz provides information mostly on lung parenchyma or thoracic cavity. Higher frequencies take the airway lumen route [12] and are thus best detected at the trachea or neck.



(a) Power spectra of normal lung sounds in a healthy adult male subject, recorded with a contact sensor over the superior right lower lobe



(b) Power spectra of normal tracheal sounds in a healthy man, recorded with a contact sensor at the suprasternal notch

Figure 2.3: Graphs from H. Pasterkamp et al. [11]. Higher Frequency content can be observed in (b) due to the direct coupling to the airway lumen.

The additional graphs using Heliox show the parenchymal pathway is not affected by the change in gas density as sound is conveyed through the tissue.

Pathological conditions can produce additional sounds as listed in Table 2.1.

Table 2.1: Categories of Respiratory Sounds - adopted from [11]

Respiratory Sound	Mechanisms	Origin	Acoustics	Relevance
Basic sounds				
Normal lung sound	Turbulent flow vortices, unknown mechanisms	Central airways (expiration), lobar to segmental airways (inspiration)	Low-pass filtered noise (range < 100 to > 1000 Hz)	Regional ventilation, airway caliber
Normal tracheal sound	Turbulent flow, flow impinging on airway walls	Pharynx, larynx, trachea, large airways	Noise with resonances (range < 100 to > 3000 Hz)	Upper airway configuration
Adventitious sounds				
Wheeze	Airway wall flutter, vortex shedding	Central and lower airways	Sinusoid (range ~ 100 to > 1000 Hz; duration typically > 80 ms)	Airway obstruction, flow limitation
Rhonchus	Rupture of fluid films, airway wall vibrations	Larger airways	Series of rapidly damped sinusoids (typically < 300 Hz; duration > 100 ms)	Secretions, abnormal airway collapsibility
Crackle	Airway wall stress-relaxation	Central and lower airways	Rapidly damped wave deflection (duration typically < 20 ms)	Airway closure, secretions

* This table lists only the major categories of respiratory sounds and does not include other sound such as squawks, friction rubs, grunting, snoring, or cough. Current concepts on sound mechanisms and origin may be incomplete and unconfirmed

2.4 Joints and Tendons

Acoustic Emission (AE) of joints under load is widely considered to be a natural phenomenon [13]. As Emissions are generated by the movement of the joint surfaces, this dynamic approach is the main advantage over static mode examinations such as MRI or Ultrasound and can be correlated to joint angle and load. By means of placing a sensor on the skin over bony prominence acoustic signals can be detected. Signals are generated by the interaction of the articular surfaces, ligaments and tendons and range up to 500 kHz [14]

In order to analyze the AE signal, it is common to describe individual hits. A hit is defined as a transient, burst like event satisfying a set of definition parameters. The most commonly used parameters are the amplitude, duration and frequency content of the hit [13]. A graphic overview is given in Fig. 2.4.

Another metric for assessing the health of cartilaginous joints is the analysis of the continuous activity summarized in the b-value [15].

$$\log_{10}(N_{AE}) = a - bA_{dB} \quad (2.1)$$

$$A_{dB} = 20 \log_{10} \left(\frac{V_{\text{peak}}}{V_{\text{ref}}} \right) \quad (2.2)$$

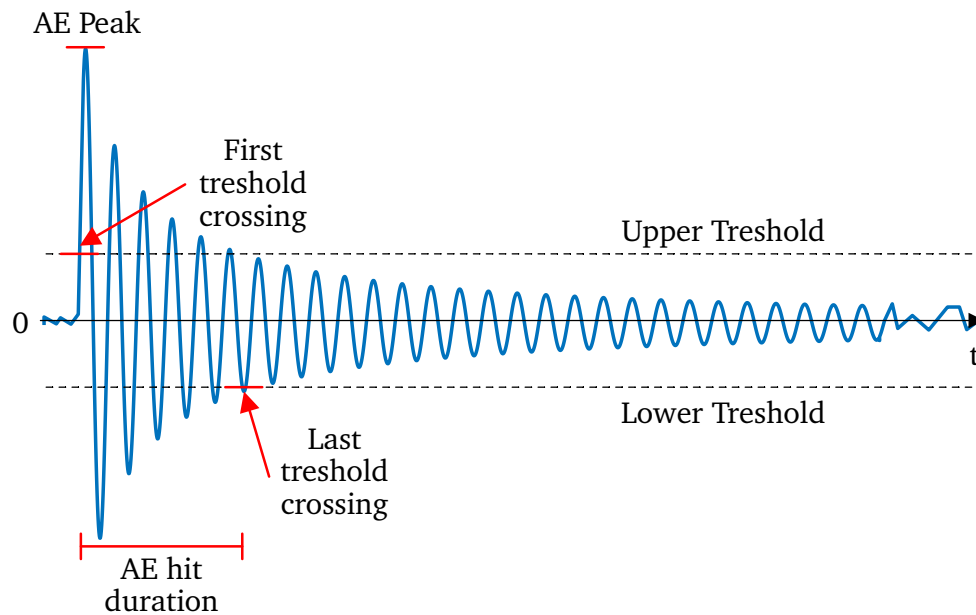


Figure 2.4: Graphic overview of the most commonly used parameters for describing and classifying an AE hit.

The b-value is derived from the field of seismology, expressing the logarithmic relationship between magnitude and frequency of a shockwave generated by an earthquake. In the context of joints, the coefficient b describes the decrease in the number of hits N_{AE} with an increasing amplitude A_{dB} serving as the detection threshold.

[15] showed, that the b-value is significantly lower in patients with an injured knee (1.46 ± 0.35) (torn anterior cruciate ligaments, torn lateral menisci, and sprained medial collateral ligaments) compared to healthy controls (1.92 ± 0.2), meaning that more high amplitude hits are present. [14] too observed a higher number of high amplitude hits in patients with osteoarthritis compared to healthy controls.

2.5 Muscle Sounds

2.6 Overview

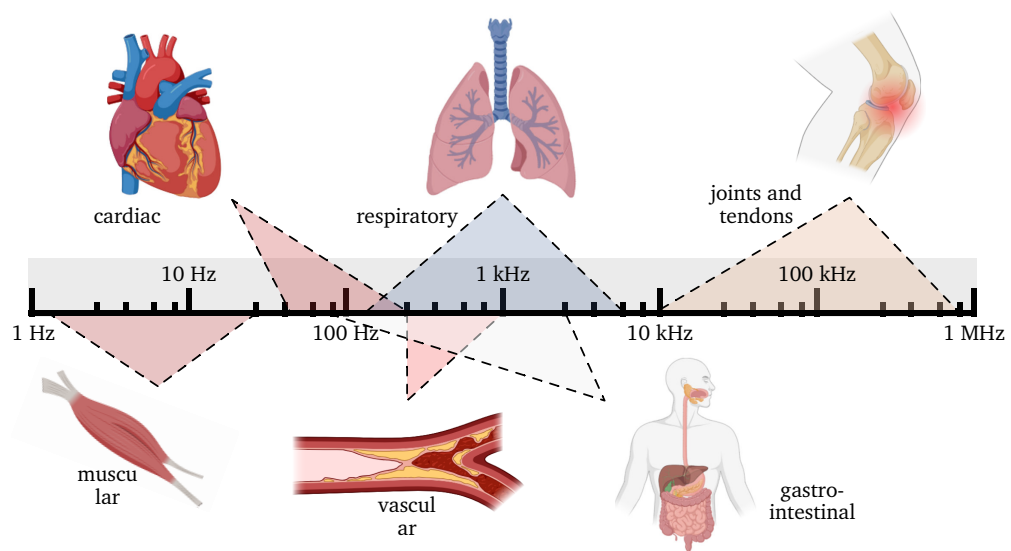


Figure 2.5: Overview of the frequency spectrum of acoustic signals. Icons from biorender.com

3 Sensor Types

A typical AE monitoring system takes its input from an AE sensor that converts dynamic surface motion into electric signals . Due to the low amplitude of AE signals and the high impedance of piezoelectric sensors, the amplification is usually carried out in two steps, firstly by the pre-amplifier and subsequently by the main amplifier. Band pass filters are also employed to eliminate unwanted noise, such as sensor and electrical circuit noise, electromagnetic interference, background and technological acoustics noise.

3.1 Low Frequency Sensors

3.1.1 Seismographic Sensors

3.2 High Frequency Sensors

3.2.1 Piezoelectric Sensors

3.3 Ferroelectret Sensors



4 Discussion



5 Conclusion

Bibliography

- [1] M. R. Montinari and S. Minelli, "The first 200 years of cardiac auscultation and future perspectives," *Journal of Multidisciplinary Healthcare*, vol. 12, pp. 183–189, Mar. 6, 2019, ISSN: 1178-2390. DOI: 10.2147/JMDH.S193904 Accessed: Apr. 17, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6408918/>
- [2] S. M. Debbal and F. Bereksi-Reguig, "Computerized heart sounds analysis," *Computers in Biology and Medicine*, vol. 38, no. 2, pp. 263–280, Feb. 1, 2008, ISSN: 0010-4825. DOI: 10.1016/j.compbimed.2007.09.006 Accessed: Apr. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010482507001771>
- [3] S. Li, F. Li, S. Tang, and W. Xiong, "A review of computer-aided heart sound detection techniques," *BioMed Research International*, vol. 2020, no. 1, p. 5846191, 2020, eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1155/2020/5846191>, ISSN: 2314-6141. DOI: 10.1155/2020/5846191 Accessed: Apr. 14, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2020/5846191>
- [4] W. Chan et al., "Haemodynamic and structural correlates of the first and second heart sounds in pulmonary arterial hypertension: An acoustic cardiography cohort study," Jan. 1, 2013. DOI: 10.1136/bmjopen-2013-002660 Accessed: Apr. 20, 2026. [Online]. Available: <https://bmjopen.bmj.com/content/3/4/e002660.full>
- [5] H. Ren, H. Jin, C. Chen, H. Ghayvat, and W. Chen, "A novel cardiac auscultation monitoring system based on wireless sensing for healthcare," *IEEE Journal of Translational Engineering in Health and Medicine*, vol. PP, pp. 1–1, Jun. 14, 2018. DOI: 10.1109/JTEHM.2018.2847329
- [6] R. L. Kirkeeide, D. F. Young, and N. R. Cholvin, "Wall vibrations induced by flow through simulated stenoses in models and arteries," *Journal of Biomechanics*, vol. 10, no. 7, pp. 431–441, Jan. 1, 1977, ISSN: 0021-9290. DOI: 10.1016/0021-9290(77)90020-3 Accessed: Apr. 20, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0021929077900203>
- [7] F. Khalili, P. T. Gamage, A. Taebi, M. E. Johnson, R. B. Roberts, and J. Mitchel, "Spectral decomposition and sound source localization of highly disturbed flow through a severe arterial stenosis," *Bioengineering*, vol. 8, no. 3, p. 34, Mar. 4, 2021, ISSN: 2306-5354. DOI: 10.3390/bioengineering8030034 Accessed: Apr. 13, 2026. [Online]. Available: <https://www.mdpi.com/2306-5354/8/3/34>
- [8] R. Ranta, V. Louis-Dorr, C. Heinrich, D. Wolf, and F. Guillemin, "Digestive activity evaluation by multichannel abdominal sounds analysis," *IEEE Transactions on Biomedical Engineering*, vol. 57, no. 6, pp. 1507–1519, Jun. 2010, ISSN: 0018-9294, 1558-2531. DOI: 10.1109/TBME.2010.2040081 Accessed: Apr. 20, 2026. [Online]. Available: <https://ieeexplore.ieee.org/document/5415618/>

-
- [9] C. Dimoulas, G. Kalliris, G. Papanikolaou, V. Petridis, and A. Kalampakas, "Bowel-sound pattern analysis using wavelets and neural networks with application to long-term, unsupervised, gastrointestinal motility monitoring," *Expert Systems with Applications*, vol. 34, no. 1, pp. 26–41, Jan. 1, 2008, ISSN: 0957-4174. DOI: 10.1016/j.eswa.2006.08.014 Accessed: Apr. 20, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417406002521>
- [10] E. Andrès, R. Gass, A. Charloux, C. Brandt, and A. Hentzler, "Respiratory sound analysis in the era of evidence-based medicine and the world of medicine 2.0," *Journal of Medicine and Life*, vol. 11, no. 2, pp. 89–106, 2018, ISSN: 1844-122X. Accessed: Apr. 20, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6101681/>
- [11] H. Pasterkamp, S. S. Kraman, and G. R. Wodicka, "Respiratory sounds," *American Journal of Respiratory and Critical Care Medicine*, vol. 156, no. 3, pp. 974–987, Sep. 1, 1997, ISSN: 1073-449X, 1535-4970. DOI: 10.1164/ajrccm.156.3.9701115 Accessed: Apr. 13, 2026. [Online]. Available: <https://academic.oup.com/ajrccm/article/156/3/974/8530878>
- [12] M. Mahagnah and N. Gavriely, "Gas density does not affect pulmonary acoustic transmission in normal men," *Journal of Applied Physiology*, vol. 78, no. 3, pp. 928–937, Mar. 1995, ISSN: 8750-7587. DOI: 10.1152/jappl.1995.78.3.928
- [13] D. C. Whittingslow, H.-K. Jeong, V. G. Ganti, N. J. Kirkpatrick, G. F. Kogler, and O. T. Inan, "Acoustic emissions as a non-invasive biomarker of the structural health of the knee," *Annals of biomedical engineering*, vol. 48, no. 1, pp. 225–235, Jan. 2020, ISSN: 0090-6964. DOI: 10.1007/s10439-019-02333-x Accessed: Apr. 20, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6930345/>
- [14] L.-K. Shark, H. Chen, and J. Goodacre, "Discovering differences in acoustic emission between healthy and osteoarthritic knees using a four-phase model of sit-stand-sit movements," *The Open Medical Informatics Journal*, vol. 4, pp. 116–125, Jul. 27, 2010, ISSN: 1874-4311. DOI: 10.2174/1874431101004010116 Accessed: Apr. 13, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3048332/>
- [15] H. K. Jeong, D. Whittingslow, and O. T. Inan, "B-value: A potential biomarker for assessing knee-joint health using acoustical emission sensing," *IEEE Sensors Letters*, vol. 2, no. 4, pp. 1–4, Dec. 2018, ISSN: 2475-1472. DOI: 10.1109/LSSENS.2018.2871981 Accessed: Apr. 21, 2026. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/8471187>